

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6CJ304 / MAT8MN304			
Course Title	COMPLEX ANALYSIS II			
Type of Course	Major			
Semester	VI			
Academic Level	300-399			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	-	60
Pre-requisites	Idea of complex numbers, Polar representations, Differentiability and Analyticity. As a Part II course, it is desirable to have the necessary details of MAT5CJ303 (Complex Analysis I) learned in advance of this course.			
Course Summary	We continue from Complex Analysis-I and begin by discussing complex integrals, followed by Cauchy-Goursat Theorem. Independence of path, Cauchy's Integral formula, sequence and series of complex numbers are next studied. It is then followed by Taylor series, Laurent series. zeros and poles, and Residue Theorem. Applications of Residue theorem are also discussed.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply the principles of real and complex integrals, including the Cauchy-Goursat theorem	Ap	P	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO2	Analyse the independence of path and evaluate the Cauchy's integral formulas, along with understanding their consequences and applications.	An	C	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO3	Create and utilize Taylor and Laurent series, and apply the residue theorem to evaluate complex functions and integrals.	C	F	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Complex Analysis (Third Edition): Dennis G. Zill & Patric D. Shanahan, Jones & Bartlett Learning, 2018.			
Module	Unit	Content	Hrs (60)	External Marks (70)
I	Module I		12	Min.15
	1	Section 5.1-Real Integrals.		
	2	Section 5.2-Complex Integrals-up to and including Example 2		
	3	Section 5.2- Complex Integrals- All the topics after Example 2		
	4	Section 5.3- Cauchy- Goursat Theorem-up to and including Example 4.		
	5	Section 5.3 -Cauchy- Goursat Theorem-All the topics after Example 4.		
II	Module II		12	Min.15
	6	Section 5.4- Independence of Path		
	7	Section 5.5 -Cauchy's Integral Formulas and Their Consequences- Cauchy's Two Integral Formulas (All the topics in 5.5.1)		
	8	Section 5.5 -Cauchy's Integral Formulas and Their Consequences- Some Consequences of the Integral Formulas (All the topics in 5.5.2)		
	9	Section 6.1 -Sequences and Series- up to and including Example 4.		
	10	Section 6.1- Sequences and Series- All the topics after Example 4.		
III	Module III		14	Min.15
	11	Section 6.2 -Taylor Series-up to and Excluding Theorem 6.2.4.		
	12	Section 6.2- Taylor Series-From Theorem 6.2.4 to Example 3.		
	13	Section 6.3 -Laurent Series-up to and including Example 1.		
	14	Section 6.3- Laurent Series- All the topics after Example 1(proof of Laurent's Theorem is optional)		
	15	Section 6.4 -Zeros and Poles- up to and including Example 2.		
IV	Module IV		10	
	17	Section 6.5 -Residues and Residue Theorem-up to and including Example 3.		
	18	Section 6.5 - Residues and Residue Theorem-All the topics after Example 3.		

	19	Section 6.6- Some Consequences of the Residue Theorem- Evaluation of Real Trigonometric Functions (up to and including example1 of 6.6.1)	Min.15
	20	Section 6.6 -Some Consequences of the Residue Theorem- Evaluation of Real Improper Integrals (up to and including Example 2)	
	21	Section 6.6 -Some Consequences of the Residue Theorem- Theorem 6.6.1 and Example 3.	
	22	Section 6.6 -Some Consequences of the Residue Theorem- Theorem 6.6.2 and Example 4.	
V	Module V (Open Ended)		12
		Definite Integrals, Line Integrals in the Plane, Indented Contours	
		Integration along a Branch Cut, The Argument Principle	
		Rouche’s Theorem and its applications	
References			
	1	Brown, James Ward, and Ruel V. Churchill. Complex variables and applications. McGraw-Hill, 2009.	
	2	Stein, Elias M., and Rami Shakarchi. Complex analysis. Vol. 2. Princeton University Press, 2010.	
	3	Burckel, Robert B. An Introduction to Classical Complex Analysis: Vol. 1. Vol. 64. Burkhouse, 2012.	
	4	Hormander, Lars. An introduction to complex analysis in several variables. Elsevier, 1973.	
	5	Priestley, Hilary A. Introduction to complex analysis. OUP Oxford, 2003.	
	6	Silverman, Richard A. Introductory complex analysis. Courier Corporation, 2013.	
	7	Bak, Joseph, Donald J. Newman, and Donald J. Newman. <i>Complex analysis</i> . Vol. 8. New York: Springer, 2010.	

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	1	0	3	0	3	0	3	0	0
CO 2	1	2	1	0	2	0	3	0	3	0	0
CO 3	1	2	1	0	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓